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Surname:															
First Name(s):															

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

June exam	1 June 2022
Time : 2 hours	Maximum marks: 62
	Full marks: 60

- This question paper consists of 8 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Basic scientific calculators also allowed.
- For Section A, only final answers will be marked. For Section B, full answers must be given.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	TOTAL
7	10	6	4	10	12	8	5	62

Student Number:

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Section A

In this section, give your final answer only. Do not show your working.

Question 1. [7 points] Let

$$A = \begin{pmatrix} 2 & 0 & 1 & 3 & -1 \\ -4 & -1 & 0 & -6 & 2 \\ 0 & 3 & -6 & 3 & 1 \\ 2 & -1 & 3 & 6 & 0 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 0 & 1 & 3 & -1 \\ 0 & -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The matrix U is a row-echelon form of the matrix A (you need not check this).

(a) Give the rank and nullity of A .

Rank 3

Nullity 2

(b) Give a basis for

(i) $NS(A)$

For instance $\left\{ \begin{pmatrix} -1 \\ 4 \\ 2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 0 \\ -1 \\ 3 \end{pmatrix} \right\}.$

(ii) $CS(A)$

For instance $\left\{ \begin{pmatrix} 2 \\ -4 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ -6 \\ 3 \\ 6 \end{pmatrix} \right\}.$

(c) Which one of the following statements is true?

- (i) The system $A\mathbf{x} = \mathbf{b}$ has a unique solution for all $\mathbf{b} \in \mathbb{R}^4$.
- (ii) The system $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions for all $\mathbf{b} \in \mathbb{R}^4$.
- (iii) The system $A\mathbf{x} = \mathbf{b}$ has no solution for some $\mathbf{b} \in \mathbb{R}^4$.
- (iv) The system $A\mathbf{x} = \mathbf{b}$ has a unique solution for some $\mathbf{b} \in \mathbb{R}^4$.
- (v) The system $A\mathbf{x} = \mathbf{b}$ has a no solution for all $\mathbf{b} \in \mathbb{R}^4$.

Give the number of the correct statement: (iii)

Student Number:								
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Question 2. [10 points] Let $A = \begin{pmatrix} 4 & -3 & -3 \\ 2 & -3 & -1 \\ 0 & 0 & -2 \end{pmatrix}$.

(a) Find all the eigenvalues of A .

$3, -2, -2$

(b) Is A invertible?

Yes

Give a reason for your answer using only your answer to (a).

A is invertible because 0 is not an eigenvalue of A .

(c) For each eigenvalue of A , find a basis for the corresponding eigenspace.

For $\lambda_1 = 3$, a basis is for instance $\left\{ \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} \right\}$

For $\lambda_2 = -2$, a basis is for instance $\left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \right\}$

(d) Is A diagonalizable? Yes

Give a reason for your answer.

A has three linearly independent eigenvectors.

(e) If A is diagonalizable, find an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$. If it is not, find a basis B for $\mathbb{R}^3$ containing at least two eigenvectors of A .

We can take for instance $P = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, $D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

Question 3. [6 points] For $p(x), q(x) \in P_2$, put $\langle p(x), q(x) \rangle = \int_0^1 p(x)q(x) dx$. You may assume that this defines an inner product on P_2 . Let $p(x) = x - 2$ and $q(x) = 2x^2$.

(a) Find $\langle p(x), q(x) \rangle$.

$-5/6$

Student Number:								
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(b) Find $\|q(x)\|$.

$$2/\sqrt{5}$$

(c) For which value of b will $r(x) = x - b$ be orthogonal to $q(x)$?

$$b = \frac{3}{4}$$

Student Number:								
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Section B

In this section, give full answers to all the questions. Show all your working.

Question 4. [4 points] Let $A = \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix}$. Write A as a product of elementary matrices.

For instance $A = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 10 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

Question 5. [10 points] The subset S of $M_{2 \times 2}(\mathbb{R})$ is given by

$$S = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 2 \\ 1 & 0 \end{pmatrix} \right\}.$$

Answer the following questions, in each case showing clearly how you arrived at your answer.

(a) Is S linearly independent?

S is not linearly independent. The last matrix is the sum of the other three matrices.

(b) Is $\text{span}(S) = M_{2 \times 2}(\mathbb{R})$?

If it is, explain why it is, or if it is not, find an equation in a, b, c and d that all matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in $\text{span}(S)$ must satisfy.

$$a - 2b + 2c + d = 0$$

(c) Write down a basis B for $\text{span}(S)$.

For instance $B = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right\}.$

(d) Is B a basis for $M_{2 \times 2}(\mathbb{R})$? Give a reason for your answer. If not, find a matrix A such that $B \cup \{A\}$ is a basis for $M_{2 \times 2}(\mathbb{R})$ and explain how you chose the matrix.

B is not a basis for $M_{2 \times 2}(\mathbb{R})$, since the dimension of $M_{2 \times 2}(\mathbb{R})$ is 4 and therefore every basis for it must have 4 elements.

For A , we can take any matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ such that $a - 2b + 2c + d \neq 0$, for instance $A = \begin{pmatrix} 1 & 1 \\ 1 & 70 \end{pmatrix}$

Student Number:								
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Question 6. [12 points] Let $B = \{x^2 + 2, x - 1, 3x - x^2\}$ and $S = \{p(x) \in P_2 : p(0) = p(2)\}$. The function $T : P_2 \rightarrow M_{2 \times 2}(\mathbb{R})$ is a linear transformation and

$$T(p(x)) = \begin{pmatrix} p'(0) & p(0) \\ p(0) & p''(0) \end{pmatrix}, \text{ for } p(x) \in P_2.$$

You may assume that B is a basis for P_2 .

- (a) Is S a subspace of P_2 ? Show how you arrived at your answer.

Yes, S is a subspace of $M_{2 \times 2}(\mathbb{R})$.

S is obviously nonempty.

Closure under addition: Suppose p and q are polynomials such that $p(0) = p(2)$ and $q(0) = q(2)$. Then $(p + q)(0) = p(0) + q(0) = p(2) + q(2) = (p + q)(2)$. Hence $p + q \in S$.

Closure under scalar multiplication: Let p be a polynomial such that $p(0) = p(2)$ and let λ be any scalar. Then $(\lambda p)(0) = \lambda p(0) = \lambda p(2) = (\lambda p)(2)$. Hence $\lambda p \in S$.

- (b) Find $T(p(x))$ if $p(x) = ax^2 + bx + c$.

$$\begin{pmatrix} b & c \\ c & 2a \end{pmatrix}$$

- (c) Find the matrix ${}_C[T]_B$ representing T with respect to the basis B in P_2 and the standard basis C for $M_{2 \times 2}(\mathbb{R})$.

$${}_C[T]_B = \begin{pmatrix} 0 & 1 & 3 \\ 2 & -1 & 0 \\ 2 & -1 & 0 \\ 2 & 0 & -2 \end{pmatrix}$$

- (d) Find the kernel of T .

The kernel of T consists of all polynomials $ax^2 + bx + c$ such that

$$\begin{pmatrix} b & c \\ c & 2a \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

which implies that $a = b = c = 0$. It follows that the kernel of T consists only of the zero polynomial.

- (e) Find a basis for the range of T .

$$\text{For instance } \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}.$$

- (f) Is T a linear isomorphism? Give a reason for your answer.

No. T is not onto.

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Question 7. [8 points] Read the following result and its proof and then answer the questions following it.

Theorem Let V be an n -dimensional vector space and $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a linearly independent subset of V . Then S is a basis for V .

Proof Suppose S is not a basis for V (1)

Then there is a $\mathbf{v} \in V$ such that $\mathbf{v} \notin \text{span } S$ (2)

Suppose $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n + \lambda_{n+1} \mathbf{v} = \mathbf{0}$ (3)

If $\lambda_{n+1} \neq 0$, then $\mathbf{v} = \frac{-1}{\lambda_{n+1}}(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n)$, (4)

a contradiction. Hence we must have $\lambda_{n+1} = 0$ (5)

If we substitute this in (3), it follows from the resulting equation that

$\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$ (6)

It follows that $S \cup \{\mathbf{v}\}$ is linearly independent. (7)

But this contradicts the fact that V is n -dimensional. (8)

Therefore S must be a basis for V .

(a) What method of proof is used here?

Proof by contradiction.

(b) Explain how line (2) follows from line (1).

We know that S is linearly independent. Hence if S is not a basis for V then it must be because S does not span V , which implies that there is a vector $\mathbf{v} \in V$ that is not the span of S .

(c) What is the contradiction in line (4)?

We assumed that $\mathbf{v} \notin \text{span } S$ but in line (4), $\mathbf{v}$ is written as a linear combination of vectors in S .

(d) Explain why we can conclude in line (6) that $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$.

If we substitute λ_{n+1} in (3) then we get the equation $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0}$. Since S is linearly independent, this implies that $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$.

(e) Why can we say in line (7) that $S \cup \{\mathbf{v}\}$ is linearly independent?

In (3), we assumed that $\lambda_1, \lambda_2, \dots, \lambda_{n+1}$ are any scalars such that $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n + \lambda_{n+1} \mathbf{v}_{n+1} = \mathbf{0}$. In (5), we have shown that it follows that $\lambda_{n+1} = 0$ and in (6) that $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$. Hence we have shown that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n, \mathbf{v}_{n+1}$ are linearly independent.

(f) Why does this contradict the fact that V is n -dimensional? (You may quote a theorem here.)

Because any set of $n + 1$ vectors in an n -dimensional vector space is linearly dependent.

(g) Let T be a subset of an n -dimensional vector space V such that $\text{span}(T) = V$, but T is not a basis of V . What can be said about the number of elements in T ? Give a reason for your answer.

Student Number:								
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There must be more than n vectors in T .

Fewer than n vectors cannot span an n -dimensional vector space. Further, if T contains exactly n vectors and spans V then T is a basis for V .

Question 8. [5 points] We will call a square matrix A *balanced* if $NS(A) = CS(A)$. Prove or disprove:

(a) There is a 2×2 balanced matrix.

This is true.

Take for instance $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then $NS(A) = CS(A) = \left\{ \lambda \begin{pmatrix} 0 \\ 1 \end{pmatrix} : \lambda \in \mathbb{R} \right\}$.

(b) There is a 3×3 balanced matrix.

This is false.

The number of columns (and hence rows) must be even because $\dim NS(A) + \dim CS(A) = Nullity(A) + Rank(A)$ equals the number of columns in A .

(c) If A is a balanced, then A cannot be diagonalised.

This is true.

Suppose A can be diagonalised and that λ is an eigenvalue of A . Then there is a non-zero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$. Since $A\mathbf{x} \in CS(A) = NS(A)$, we have $\mathbf{0} = A(A\mathbf{x}) = A(\lambda\mathbf{x}) = \lambda A\mathbf{x} = \lambda^2\mathbf{x}$. But $\mathbf{x}$ is non-zero, and therefore $\lambda^2 = 0$, giving $\lambda = 0$. It follows that $\lambda = 0$ is the only eigenvalue of A . The assumption that A can be diagonalised gives an invertible matrix P and the diagonal matrix $D = \mathbf{O}$ such that $A = P^{-1}DP = P^{-1}\mathbf{O}P = \mathbf{O}$, which contradicts the fact that A is balanced.